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A macrophage mannose receptor 1 with dual CTLDs played a crucial role in defense against *Ecytonucleospora hepatopenaei* in *Litopenaeus vannamei*

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ABSTRACT

Mannose receptor (MR), as a crucial member of the C-type lectins (CTLs), plays an important role in the innate immune responses of invertebrates. In *Litopenaeus vannamei*, we identified a novel MR, macrophage mannose receptor 1 (designated as LvMMR1), and conducted bioinformatic analyses and functional investigations. The open reading frame (ORF) of LvMMR1 is 1008 bp, encoding 335 amino acid residues. The deduced amino acid sequence of LvMMR1 contained a signal peptide, and two C-type lectin domains (CTLDs) with special characteristic motifs, QPD, LND, EPQ, and FRD, respectively. Sequence alignment and homology analysis indicated that LvMMR1 exhibited high similarity to MRs from *Penaeus monodon* and *P. chinensis*. Tissue distribution analysis showed that LvMMR1 transcripts were widely expressed in eight examined tissues, with relatively higher expression levels in the hepatopancreas and hemocytes. Upon infection with different loads of *Ecytonucleospora hepatopenaei* (EHP), LvMMR1 expression was significantly up-regulated in hepatopancreas. *In vitro* conditions, the purified recombinant protein of LvMMR1 (designated as rLvMMR1) was able to agglutinate rabbit erythrocytes and various microorganisms with a Ca^{2+}-dependent manner, including two Gram-negative bacteria (*Escherichia coli* and *Vibrio parahaemolyticus*), two Gram-positive bacteria (*Bacillus subtilis* and *Staphylococcus aureus*), and a fungus (*Pichia pastoris*). Furthermore, six carbohydrates (D-lactose, D-xylose, maltose, D-galactose, D-mannose, and lipopolysaccharide) could effectively bind to rLvMMR1 and inhibited its agglutination activity with a dose-dependent manner. These results collectively suggested that LvMMR1 played a crucial role in the immune response of shrimp against EHP infection, and contributed to a better understanding of the immune functions of the MR family in shrimp.

1. Introduction

Litopenaeus vannamei is currently the most widely cultured shrimp species with commercial importance. However, disease outbreaks occurred frequently under intensive farming conditions, leading to substantial economic losses to the shrimp industry [1,2]. Among all the shrimp diseases, hepatopancreatic microsporidiosis (HPM) is one of the most prevalent [3]. *Ecytonucleospora hepatopenaei* (EHP) is the pathogen of HPM, which is an intracellular parasitic, single-cell eukaryotic microorganism [4,5]. EHP has a unique strategy for infection and survival, which relies on the host for ATP synthesis and disrupts the normal nutrient metabolism of host, resulting in symptoms such as growth retardation, lethargy, soft shells, and white feces [6]. Notably, EHP-infected shrimps became more susceptible to co-infections by other pathogens, such as *Vibrio parahaemolyticus*, white spot syndrome virus (WSSV), infectious hypodermal and haematopoietic necrosis virus (IHHNV), and hepatopancreatic parvovirus (HPV) [7,8]. The concurrent harms manifested during EHP infection significantly increased

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aquaculture risks. Nevertheless, the molecular pathogenesis of EHP remained unclear, and no specific therapeutic agents are currently available for EHP control [9]. Shrimps rely on the innate immune system to recognize and respond to pathogenic invasion [10,11]. Pattern recognition receptors (PRRs) are key components of the innate immune system that trigger the activation of host immune defenses by recognizing pathogen-associated molecular patterns (PAMPs) derived from viruses, bacteria, and fungi [12]. Research on components of shrimp innate immune system will contribute to a better understanding of defense mechanisms against EHP.

C-type lectins, as a classic class of pattern recognition receptors (PRRs), have been identified in various invertebrates. Their relevance for pathogen clearance has been demonstrated by *in vitro* overexpression, functional verification, and RNA silencing *in vivo* [13–15]. The mannose receptor (MR), a member of the CTL superfamily, plays a significant role in the innate immune responses [16]. According to its protein domains, MR contained a N-terminal cysteine-rich (CR) domain, a fibronectin type II (FN II) domain and eight tandem C-type lectin-like domains (CTLDs) [16]. Each CTLD contained one or more characteristic carbohydrate-recognition motifs. In vertebrates, these were often QPD and EPN motifs, whereas greater diversity existed in invertebrates [17]. These motifs were associated with the specificity and affinity of carbohydrate binding [18]. MRs recognized and bound to mannose, *N*-acetyl-glucosamine, and fucose residues on microbial surfaces by their CTLD regions, thereby initiating immune responses [19–21]. Consequently, MR was considered as an important PRR and a growing number of studies were investigating its function during host response to pathogenic infection.

Currently, multiple MRs have been cloned from invertebrates, and their roles in mediating antiviral and antibacterial activities have been highlighted [22]. For instance, RpMR1, RpMR2, RpMR3, and RpMR4 identified in *Ruditapes philippinarum* exhibited peak expression at 72 h post-infection with *V. anguillarum*. Furthermore, knockdown of RpMR1 significantly suppressed the expression of TLR4, further suggesting the involvement of MR in mediating immune processes against *Vibrio* infection [23]. Previously, a *Procambarus clarkii* MR (PcMR) containing 14 CTLDs was shown to agglutinate bacteria in a Ca^{2+}-dependent manner. Studies involving recombinant protein injection and siRNA silencing indicated that PcMR participates in the antibacterial defense of crayfish [24]. Subsequently, a novel MR (PvMR2) cloned from shrimp containing two canonical CTLDs, was found to respond to *V. anguillarum* infection. And *in vitro*, it could bind to *V. anguillarum*, *V. parahaemolyticus*, *Escherichia coli*, and *Aeromonas veronii* in a Ca^{2+}-dependent manner, reduce the adhesion capacity of harmful intestinal bacteria, enhance the adhesion of beneficial bacteria, and promote phagocytosis by hemocytes [25]. Another MR from *L. vannamei* containing a FN II domain (LvMR) was reported to agglutinate microbes in a calcium-dependent manner, and its knockdown in *V. parahaemolyticus*-infected shrimp led to increased activity of reactive oxygen species (ROS) and inducible nitric oxide synthase (iNOS), as well as up-regulated expression of most NF-κB pathway genes and inflammatory genes [26]. *Pv*MR has also been observed to participate in antiviral immunity in *L. vannamei*, which could bind to IHHNV-like particles and trigger downstream immune effects, including the upregulation of antimicrobial peptides (AMPs), such as penaeidin 3 and crustin [27]. Collectively, these studies demonstrated that MRs could recognize a broad spectrum of pathogens, and were closely associated with host immune responses. Therefore, the function of MRs in shrimp defense against pathogenic infection was worth further investigation.

In our previous study, we analyzed transcriptomics of the *L. vannamei* hepatopancreas with varying EHP loads (NCBI biological project number: PRJNA1278917), and observed that a macrophage mannose receptor 1 (GenBank accession number: XM_070114845.1, designated as LvMMR1) showed significant differential expression (4.22-fold), suggesting it was a potential candidate gene for shrimp resistance of EHP [28]. In order to clarify the function and importance of MR in shrimp innate immunity, we analyzed the domain, evolution and amino acid sequence characteristics of LvMMR1, and examined its tissue distribution and expression pattern. Functional analyses, including agglutination activity of bacteria and rabbit erythrocytes, and carbohydrate binding activity *in vitro*, revealed that LvMMR1 exhibited typical CTL functional activity. This study provided a basis for further understanding the role of MR in the immunologic mechanism of *L. vannamei*.

2. Materials and methods

2.1. Sequence and phylogenetic analysis of LvMMR1

The LvMMR1 (GenBank accession number: XM_070114845.1) was identified from prior transcriptomics analyses of EHP-infected shrimp [28]. The cDNA sequence and deduced amino acid sequences of LvMMR1 were analyzed using DNAMAN v8.08 software. The theoretical isoelectric point (pI) and molecular weight were predicted using the ExPASy Molecular Biology Server (http://web.expasy.org/Compute_pi). Putative functional domains and transmembrane regions were predicted via the Simple Modular Architecture Research Tool (SMART, http://smart.embl-heidelberg.de). To identify the deduced amino acid sequences of LvMMR1 with biological similarity from other species, BLAST searches were performed against the NCBI non-redundant protein database. Multiple sequence alignment was conducted using ClustaW2 program (http://www.ebi.ac.uk/clustalw/), and a neighbor-joining (NJ) phylogenetic tree was constructed using MEGA v12.1 software, while the reliability was assessed using 1000 bootstrap replications.

2.2. Total DNA, RNA extraction and cDNA synthesis of shrimp samples

A total of 74 shrimps with size of 13 - 16 cm were collected under the same culture conditions within the same aquaculture farm. Based on the previously established detection method and standard curve for the small subunit ribosomal gene of EHP (EHP-SSU) in our laboratory [28,29], pathogen detection performed on the shrimp hepatopancreas revealed that the shrimps included 65 naturally infected shrimps and 9 shrimps without EHP infection. And we calculated the EHP loads of the shrimp hepatopancreas and categorized the samples into four groups, including very low loads group (VL, 10^3 - 10^4 copies), medium low loads group (ML, 10^4 - 10^5 copies), medium high loads group (MH, 10^5 - 10^6 copies), and very high loads group (VH, 10^6 - 10^7 copies) [28,29]. Each sample was analyzed with three technical replicates. Total DNA was extracted from hepatopancreas of shrimp samples using nucleic acid releasing agent (NR201, GenDx, China). The RNA was extracted from eight tissues using the TRIzol Reagent (15596026CN, Thermo Fisher Scientific, USA). The concentration and quality of total DNA and RNA samples were determined by a NanoDrop One spectrophotometer (Thermo Fisher Scientific, USA) and 1.5 % agarose gel electrophoresis. Total RNA was reversely transcribed into cDNA using HiScript III All-in-one RT SuperMix Perfect for qPCR (R333, Vazyme, China), and stored at −20 °C.

2.3. Expression profiling of LvMMR1

The expression level of LvMMR1 was detected using SYBR Green-based qPCR to analyze its tissue distribution in healthy shrimps and its expression changes in four groups with different EHP loads, including VL, ML, VH, MH groups. Two specific primers, LvMMR1-qF and LvMMR1-qR, were used to detect LvMMR1 expression levels, with EF-1α serving as the internal reference gene. Primer sequences were listed in Table 1. The reaction system consists of 10 μL of 2 × ChamQ SYBR qPCR Master Mix (Q311-02, Vazyme, China), 1 μL of each primer (10 mM), 1 μL of cDNA template, and 7 μL of RNase-free ddH_2O. The amplification protocol consisted of an initial step at 95 °C for 30 s, followed by 39

Table 1
Information of the primers used in this study.

Primers	Sequence (5′-3′)	Tm value	Amplicon length
Clone primers			
LvMMR1-cF	ATGTTTCCGTATTTGGCATCAGCTT	65.4 °C	1005 bp
LvMMR1-cR	CCCACGATAAACACACAGAGGG	61.8 °C	
qPCR primers			
LvMMR1-qF	ATCACTTCGCCAACGCAGTT	60.5 °C	86 bp
LvMMR1-qR	GAGCCGTCCAGCCAGAATTG	62.8 °C	
EF-1α-qF	TGGCTACTCACCTGTGCTTG	56.8 °C	100 bp
EF-1α-qR	CCAGCTCCTTACCAGTACGC	57.5 °C	

cycles of 95 °C for 10 s and 60 °C for 30 s, followed by a dissociation step was performed at the end of each cycle to verify amplification specificity. The relative expression of LvMMR1 was calculated using the $2^{-\Delta\Delta Ct}$ method [30]. All data were presented as the mean ± standard deviation of the relative mRNA level of LvMMR1. Statistical analysis was conducted using one-way ANOVA followed by Duncan's multiple range test in GraphPad Prism v8.3.0 software, with a p value < 0.05 considered statistically significant.

2.4. Protein expression and purification of recombinant LvMMR1

The coding sequence (CDS) of LvMMR1 was amplified using specific primers LvMMR1-cF and LvMMR1-cR, which contained the restriction enzyme sites, Nco I and Xho I. The amplified product was ligated into the pET-28a (+) vector. The recombinant plasmid was then transformed into *E. coli* BL21 (DE3) cells (CD701, TransGen, China) for the prokaryotic expression of the recombinant protein (rLvMMR1). In prokaryotic system, the expression of rLvMMR1 was induced by isopropyl β-D-1-thiogalactopyranoside (IPTG) at 37 °C for 4 h. The rLvMMR1 was expressed as inclusion bodies with the His-tag at N-terminal, yielding a molecular weight of approximately 37 kDa. The protein was subsequently purified using a PD-10 desalting column pre-packed with Sephadex G-25 resins (17085101, Cytiva, USA) and Ni^{2+} chelating Sepharose column (17371202, Cytiva, USA). The purified rLvMMR1 was concentrated using ultrafiltration tubes (UFC901096, Millipore, USA). The purity and integrity of the final protein preparation rLvMMR1 were analyzed by 12 % sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE), and the purified protein was used for subsequent functional assays.

2.5. Agglutination assays of rLvMMR1

The agglutination activity of recombinant LvMMR1 protein (rLvMMR1) as a CTL was preliminarily assessed using rabbit erythrocytes (MP20025, Yuanye, China). Briefly, the erythrocytes were collected by centrifugation at 4000×*g* for 5 min and washed three times with 2 % TBS. The rabbit erythrocytes were resuspended in 2 % TBS or 2% TBS with 5 mM $CaCl_2$ (2 % TBS-Ca^{2+}) buffer. To evaluate the agglutination activity of rLvMMR1 and its Ca^{2+} dependence, rLvMMR1 (0 - 2 mg/mL) was incubated with the rabbit erythrocytes suspensions in different buffer conditions, including TBS, TBS-Ca^{2+}, and TBS-Ca^{2+} with 10 mM EDTA (B540625, Sangon, China). BSA incubated with rabbit erythrocytes resuspended in TBS-Ca^{2+} buffer served as the negative control. All mixtures were prepared in a 96-well V-bottom plate, and incubated at room temperature for 1 h. Agglutination was subsequently observed under a fluorescence inverted microscopy (Axio Observer 7, Carl Zeiss, Germany).

The agglutination activity of rLvMMR1 *in vitro* was investigated using two Gram-negative bacteria (*E. coli* and *V. parahaemolyticus*), two Gram-positive bacteria (*Bacillus subtilis* and *Staphylococcus aureus*) and fungi (*Pichia pastoris*). Briefly, five bacteria cultured to the logarithmic growth phase (10^9 CFU/mL) were collected, and resuspended in 2 % TBS or 2 % TBS-Ca^{2+} buffer. Then, these bacteria were labeled with fluorescein isothiocyanate (FITC). To determine whether the agglutination function of rLvMMR1 was Ca^{2+}-dependent pattern, 2 mg/mL of rLvMMR1 was incubated with the microbial suspensions in different buffer conditions, including TBS, TBS-Ca^{2+}, and TBS-Ca^{2+} with 10 mM EDTA (B540625, Sangon, China). Only bacteria resuspended in TBS-Ca^{2+} buffer served as the blank control, and BSA incubated with bacteria in TBS-Ca^{2+} buffer served as the negative control. The 10 μL of each mixture was placed on glass slides, and observed under a fluorescence microscopy (BX53, Olympus, Japan).

2.6. Carbohydrate binding assays of rLvMMR1

To further explore the binding types and binding strength between rLvMMR1 and PAMPs, we selected three monosaccharides (D-xylose, D-mannose, D-galactose), two disaccharides (D-lactose, maltose), and a polysaccharide (lipopolysaccharide (LPS)) were used for binding experiments *in vitro*. Carbohydrate solutions with different concentration gradients (25 mM, 50 mM, 100 mM, 200 mM) were incubated with 2 mg/mL rLvMMR1 at room temperature for 1 h, and mixed with FITC labeled bacterial solution resuspended in TBS-Ca^{2+} buffer, and observed for changes of aggregation effect using fluorescence microscopy (BX53, Olympus, Japan).

3. Results

3.1. Characterization of LvMMR1

The LvMMR1 was identified from the transcriptomic analysis of *L. vannamei* hepatopancreas infected with EHP. The full-length cDNA of LvMMR1 is 1298 bp, containing an 88 bp 5′-untranslated region (UTR) and a 202 bp 3′-UTR. The open reading frame (ORF) of LvMMR1 is 1008 bp, encoding a putative polypeptide of 335 amino acid residues (aa). The predicted polypeptide included a signal peptide (1 - 20 aa), and two C-type lectin-like domains (CTLDs) (36 - 169 aa and 199 - 332 aa) (Fig. 1A). The calculated molecular weight of the encoded protein is approximately 37.03 kDa, with a theoretical isoelectric point (pI) of 4.57. The first CTLD contains six conserved cysteine residues (Cys^{26}, Cys^{47}, Cys^{65}, Cys^{144}, Cys^{160}, Cys^{168}) and two Ca^{2+}-binding characteristic motif (QPD motif, Gln^{134} - Pro^{135} - Asp^{136}; LND motif, Leu^{155} - Asn^{156} - Asp^{157}). Similarly, the second CTLD also possesses six conserved cysteine residues (Cys^{199}, Cys^{210}, Cys^{228}, Cys^{307}, Cys^{323}, Cys^{331}) and one Ca^{2+}-binding motif (EPQ motif, Glu^{298} - Pro^{299} - Gln^{300}; FRD motif, Phe^{318} - Arg^{319} - Asp^{320}) (Fig. 1A and B).

3.2. Homologous and phylogenetic analysis of LvMMR1

Combined the BLAST results and multiple sequence alignment revealed that LvMMR1 shared a similar structure with other members of the MR family from shrimps. It exhibited high sequence identity with MRs from *P. monodon* (82.09 %), *P. chinensis* (80.9%), *P. indicus* (80.6%), *P. merguiensis* (78.81%), and *P. japonicus* (74.11%) (Fig. 1B), indicating that MRs were conserved across different shrimp species. Phylogenetic analysis demonstrated that LvMMR1 was evolutionarily closely related to the MRs of *P. chinensis* and *P. monodon*, as well as to CTLs from *P. japonicus*, *P. semisulcatus*, and *P. merguiensis* (Fig. 2).

3.3. Tissue expression and immune responses of LvMMR1

The expression level of LvMMR1 in healthy shrimp tissues was detected using the SYBR Green-qPCR. The expression level of LvMMR1 in hepatopancreas was the highest, exhibiting a Log_2(fold change) of approximately 12.19 compared to that in the intestine. The expression level of LvMMR1 in hemocytes and heart was relatively high, while its expression level in muscle, gill and intestine was relatively low (Fig. 3A).

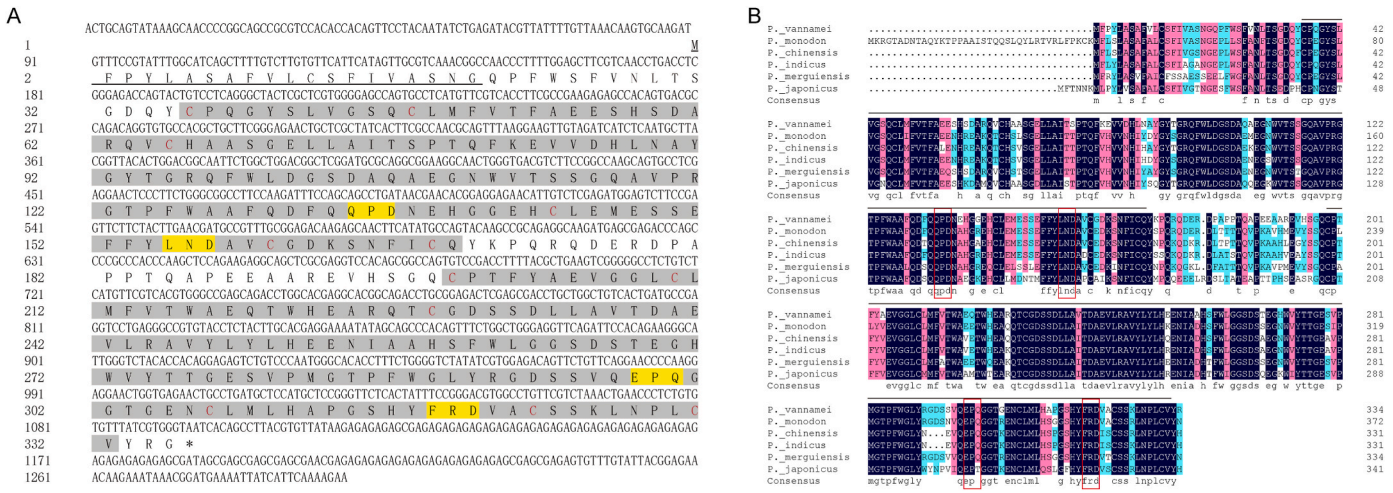


Fig. 1. Sequence characterization and multiple sequence alignment of the LvMMR1. (A) The cDNA and amino acid sequence of LvMMR1. Two CTLDs were shaded with gray. The signal peptide was underlined. The characteristic motifs for carbohydrate-binding were highlighted. The conserved cysteine residues were in yellow. (B) Multiple sequence alignment of LvMMR1 with other MRs. The used sequences were LvMMR1 (XP_069970946.1), *Penaeus monodon* (XP_037799588.1), *Penaeus chinensis* (XP_047486284.1), *Penaeus indicus* (XP_063599789.1), *Penaeus merguiensis* (AGS42194.1), *Penaeus japonicus* (XP_042878748.1). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

As shown in Fig. 3B, after EHP challenge, the expression of LvMMR1 was significantly up-regulated in the shrimp hepatopancreas with different loads compared to the NC group. The highest expression peak was observed in the MH group, with an increase of approximately 15.25-fold (p value < 0.05). The expression of LvMMR1 were significantly up-regulated approximately 4.90, 1.93, and 4.53-fold (p value < 0.05) in the VL, ML, and VH groups, respectively. These results indicated that LvMMR1 was involved in the immune defense against EHP with the hepatopancreas of shrimp infected with varying infection degrees.

3.4. Prokaryotic expression and purification of rLvMMR1

To investigate the function of LvMMR1, the recombinant plasmids were induced and overexpressed at 37 °C in *E. coli* system. SDS-PAGE analysis confirmed that rLvMMR1 was highly expressed inclusion bodies form in the pellet fraction (Fig. 4, Lane 3), exhibiting a molecular weight consistent with the predicted size of 37.03 kDa. The rLvMMR1 was subsequently purified stepwise using Ni^{2+} chelating Sepharose column. Ultimately, the purified rLvMMR1 was concentrated to a final concentration of 2 mg/mL for subsequent functional experiments (Fig. 4, Lane 6).

3.5. Hemagglutination activity of rLvMMR1

To preliminarily assess whether the purified rLvMMR1 possesses agglutination activity, rabbit erythrocytes were incubated with different concentrations of rLvMMR1 and observed for agglutination. As shown in Fig. 5A, agglutination was observed only in the mixture containing rLvMMR1 and rabbit erythrocytes resuspended in TBS-Ca^{2+} buffer, which indicated that the agglutination function of rLvMMR1 was strictly dependent on the presence of Ca^{2+}. Furthermore, a slight agglutination effect began to appear when the concentration of rLvMMR1 reached 0.05 mg/mL (Fig. 5B). The agglutination was completely inhibited with EDTA (Fig. 5B).

3.6. Agglutinating activity of rLvMMR1 to microorganisms

To better observe microbial agglutination, based on the results of hemagglutination assay, 2 mg/mL rLvMMR1 showed significant agglutination activity, which was selected for further experiments. As shown in Fig. 6, the rLvMMR1 exhibited agglutination activity against *E. coli*, *B. subtilis*, and *P. pastoris*, with the strongest agglutination observed for *B. subtilis*. In contrast, relatively weaker agglutination effects were appeared for *V. parahaemolyticus* and *S. aureus*. No agglutination was observed in both TBS + Ca^{2+}, BSA + Ca^{2+}, and rLvMMR1 groups. The agglutination effects of rLvMMR1 to microorganisms were observed only in the presence of Ca^{2+}.

3.7. Binding activity of rLvMMR1 to carbohydrates

The binding ability of rLvMMR1 to six kinds of carbohydrates was verified in this study. The rLvMMR1 was preincubated with carbohydrates and then its agglutination effects with *B. subtilis* were observed. The results showed that the binding ability of rLvMMR1 to six kinds of carbohydrates was dose-dependent as shown in Fig. 7. When the concentration of D-lactose, D-xylose, Maltose, D-galactose, and LPS reached 200 mM, the agglutination ability of rLvMMR1 to *B. subtilis* was significantly inhibited. D-mannose exhibited significant inhibition at 50 mM and almost completely inhibited the agglutination ability of rLvMMR1 at concentrations up to 200 mM.

4. Discussion

The MR, as a member of the CTL superfamily, served as an important PRR [31]. In vertebrates, MR was present in a variety of immune cells, such as macrophages and dendritic cells, that recognized surface glycoproteins of pathogens and triggered immune responses from host cells [32]. Vertebrate MRs shares a conserved domain architecture, consisting of a short cytoplasmic region, a transmembrane domain, and an extracellular region that includes a CR domain, a FN II domain, and multiple tandem CTLDs [33]. In invertebrates, an increasing number of MRs have been identified and were closely associated with innate immune responses [34,35]. In contrast, invertebrate MRs often lack one or both of the CR and FN II domains. The LvMMR1 protein identified in this study contained two CTLDs but lacked both CR and FN II domains, and this result was consistent with the previously reported PvMR2 from *L. vannamei* [25]. It is somewhat similar to PcMR identified in *P. clarkii*, but PcMR contained 14 CTLDs [36]. Another MR reported in *L. vannamei*, LvMR, contained two CTLDs along with an FN II domain [26]. Multiple sequence alignment and phylogenetic analysis confirmed the high sequence similarity and close evolutionary relationship of LvMMR1 with other invertebrate MR. MRs recognized invading

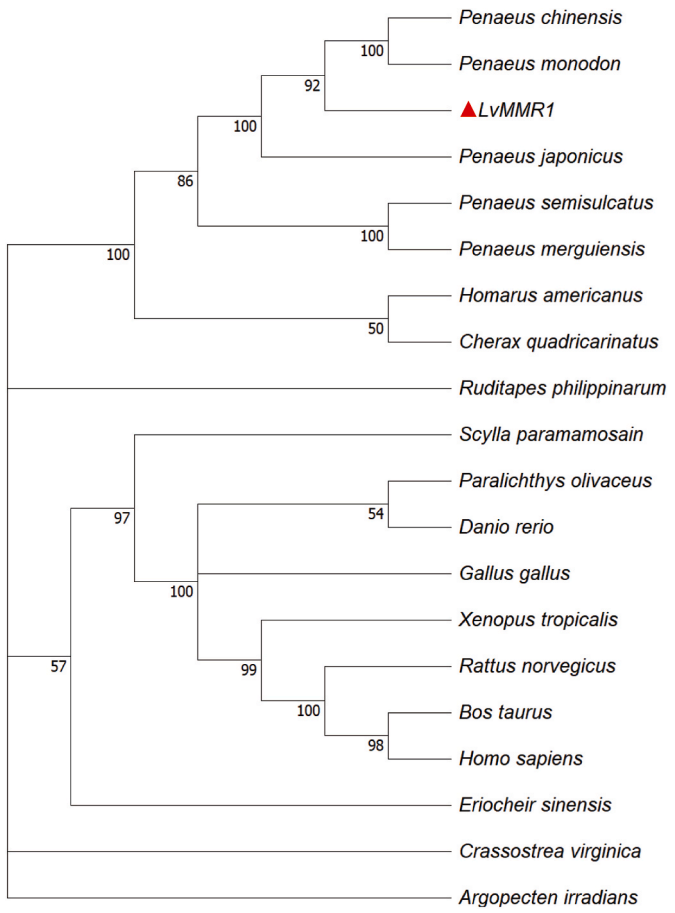


Fig. 2. Phylogenetic analysis of LvMMR1. The percentage bootstrap values were shown at the branches. Bootstrap values were calculated from 1000 replicates. LvMMR1 are shown with ▲. The sequences and their accession numbers are as follows: LvMMR1 (XP 069970946.1); *Penaeus chinensis* (XP 047486284.1*Penaeus monodondon* (XP 037799590); *Penaeus japonicus* (AGW27416.1*Penaeus semisulcatustus* (ABl97372.1); *Penaeus merguiensis* (ACR56805.1*Homarus americanusnus* (KAG7158888.1*Cherax quadricarinatustus* (XP 069953087.1*Eriocheir sinensissis* (XP 050725050.1*Scylla paramamosainain* (XP 063839877.1*Crassostrea virginicaica* (XP 022305689.1*Argopecten irradiansans* (XP 069136265.1*Ruditapes philippinarumrum* (XP 060602932.1*Gallus galluslus* (NP 001305362.1*Paralichthys olivaceuseus* (XP 069369119.1); *Xenopus tropicalis* (XP 012821081.2*Bos taurusrus* (XP 003586820.2*Rattus norvegicuscus* (NP 001099593.2*Homo sapiensens* (NP 002429.1*Danio reriorio* (NP 001410655.1).

pathogens by binding to terminal mannose, *N*-acetyl-glucosamine, and fucose residues on microorganism surfaces via their CTLD regions, thereby initiating innate immune responses [19]. These findings suggest that LvMMR1 was a novel invertebrate MR, and its involvement in the immune response during EHP infection in shrimp was closely associated with the function of two CTLDs. Each CTLD of LvMMR1 contained two conserved Ca^{2+}-binding motif, QPD, LND, EPQ, and FRD respectively, which were critical for the carbohydrate-binding specificity of LvMMR1 [37]. Carbohydrate-binding motifs exhibited sequence diversity among invertebrates, and variant motifs have been identified in other CTLs, such as EPD, EPK, EPS, EPQ, QAP, QPG, QPT, QPN, LND, and FRD [38]. Similarly, MRs in invertebrates also contained diverse carbohydrate-binding motifs, including EPN, EPS, and FND [26,39]. These variation in motifs might alter the specificity of ligand recognition and binding of MRs, potentially influencing the host's defense against different pathogens [40,41].

To further investigate the function of MR, we analyzed the

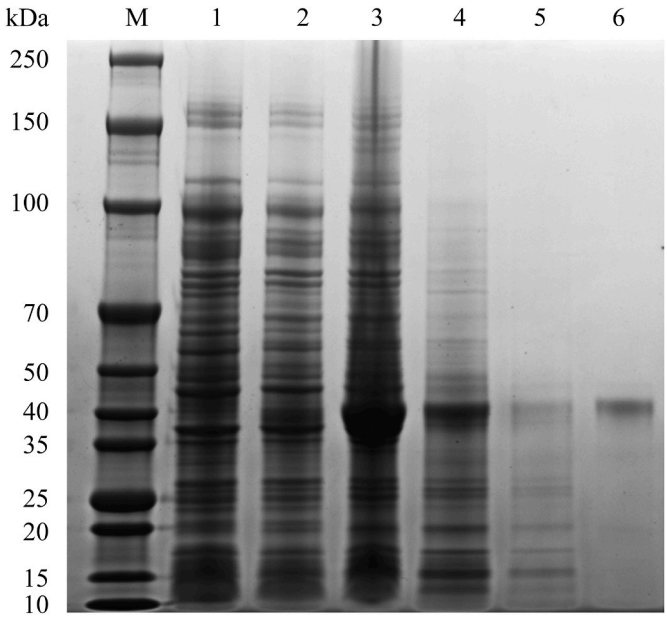


Fig. 4. SDS-PAGE analysis of rLvMMR1 expression and purification. M, protein molecular standard; Lane 1, total bacterial suspension before IPTG induction; Lane 2, total bacterial suspension after IPTG induction; Lane 3, pellet fraction after sonication; Lane 4, flow-through (FT) sample from the Ni^{2+} chelating Sepharose column; Lane 5, proteins eluted with 40 mM imidazole; Lane 6, rLvMMR1 eluted with 200 mM imidazole.

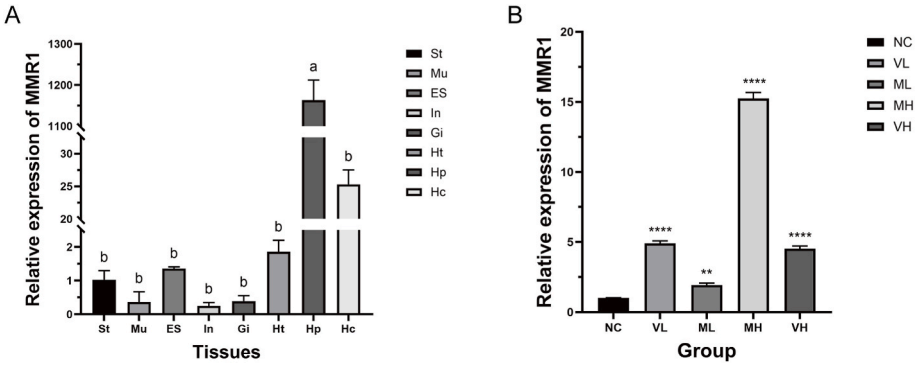


Fig. 3. The expression profile of LvMMR1. (A) The relative mRNA expression levels of LvMMR1 in different tissues of *L. vannamei*, including stomach (St), muscle (Mu), eyes (Es), intestine (In), gills (Gi), heart (Ht), hepatopancreas (Hp), and hemocytes (Hc). (B) The relative mRNA expression levels of LvMMR1 in the shrimps hepatopancreas of very low (VL, 10^3 - 10^4 copies), medium low (ML, 10^4 - 10^5 copies), medium high (MH, 10^5 - 10^6 copies), and very high (VH, 10^6 - 10^7 copies) groups.

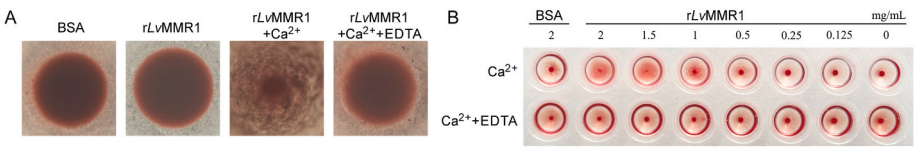


Fig. 5. The hemagglutination activity of rLvMMR1. (A) The hemagglutination effect of LvMMR1(2 mg/mL) was observed in BSA, rLvMMR1, rLvMMR1+Ca^{2+}, rLvMMR1+Ca^{2+}+EDTA groups. (B) The hemagglutination activity of LvMMR1 (0 - 2 mg/mL) was observed in BSA, rLvMMR1, rLvMMR1+Ca^{2+}, rLvMMR1+Ca^{2+}+EDTA groups.

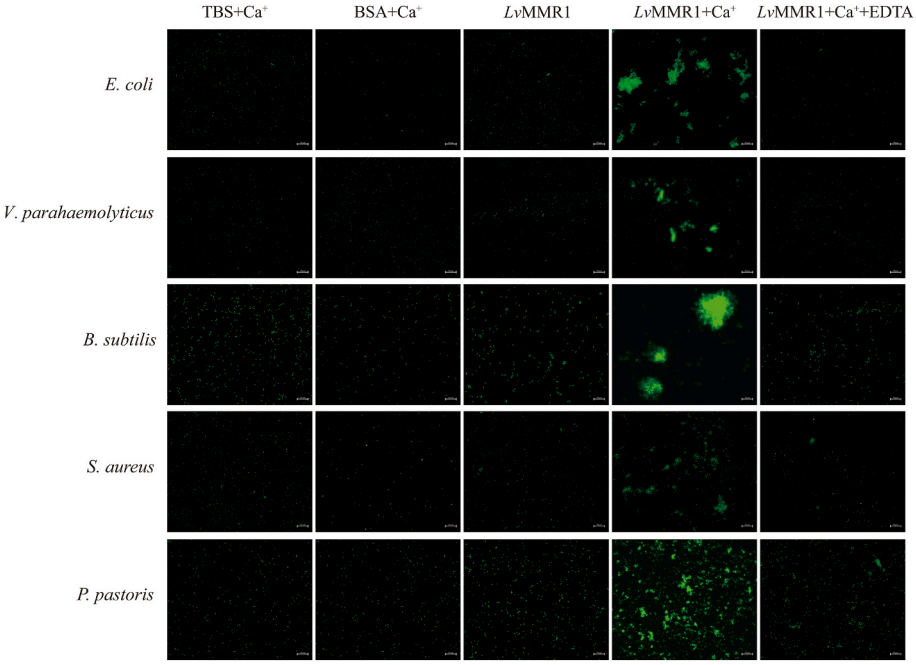


Fig. 6. The agglutination activity of rLvMMR1. The agglutination effects on Gram-negative bacteria (*Escherichia coli* and *Vibrio parahaemolyticus*), Gram-positive bacteria (*Bacillus subtilis* and *Staphylococcus aureus*) and fungi (*Pichia pastoris*), were observed in TBS + Ca^{2+}, BSA + Ca^{2+}, rLvMMR1, rLvMMR1+Ca^{2+}, rLvMMR1+Ca^{2+}+EDTA groups.

expression pattern and immune characteristics of LvMMR1 after EHP infection. Previous studies have demonstrated that down-regulation of MR could disrupt immune signaling pathways, suppressed the production of inflammatory factors, and impaired the host's antibacterial response [16]. The role of MR in shrimp immunity has also gradually obtained attention. Upon EHP infection, shrimp activated a large number of immune-related genes and pathways, including significantly up-regulated astakine, lysozyme, NACHT, LRR and PYD domains-containing protein 3 (NLRP3), and MMR [42]. Based on integrated metabolomics, transcriptomics, and microbiomics analysis of EHP-infected shrimp, MMR and betaine-homocysteine S-methyltransferase (BHMT) were prioritized as major candidate genes for EHP-resistant shrimp breeding [43]. According to transcriptomics of shrimp resistance to EHP, we identified LvMMR1 as being significantly up-regulated (4.22-fold), further focusing the important role of MR in the shrimp immune response [28]. In this study, tissue-specific expression analysis revealed that LvMMR1 was mainly expressed at high levels in the hepatopancreas and hemocytes, while detectable expression also observed in other tissues. Furthermore, after EHP challenge, LvMMR1 showed a significant up-regulation in the hepatopancreas with different EHP loads, which indicated its involvement in the shrimp immune response during EHP infection. This expression pattern is consistent with previous observations in *R. philippinarum, P. clarkii, P. trituberculatus, Fenneropenaeus merguiensis*, where MRs were also up-regulated upon various pathogenic infections [23,24,39,44]. Additionally, functional studies on PcMR in *P. clarkii* demonstrated PcMR was closely associated with antibacterial capacity of host, and knockdown of PcMR reduced the host's ability to clear bacteria [24]. And in *V. parahaemolyticus*-infected shrimp, knockdown of LvMR led to increased levels of reactive oxygen species (ROS) and inducible nitric oxide synthase (iNOS) activity in the hepatopancreas, along with enhanced expression of most NF-κB pathway genes, inflammatory genes, and redox-related genes [26]. Collectively, these studies indicated that MR could participate in the immune responses of invertebrates during pathogenic infection.

To date, numerous crustacean CTLs have been investigated and demonstrated to possess antibacterial binding and agglutination capabilities. For instance, PcNTC from *P. clarkii* was shown to bind to both Gram-positive bacteria (*S. aureus*, *B. subtilis*, and *B. thuringiensis*) and Gram-negative bacteria (*E. coli*, *V. parahaemolyticus*, and *A. hydrophila*) [36]. Similarly, SpCTL-E and SpCTL-F identified in *Scylla paramamosain* were reported to agglutinate Gram-negative bacteria and fungi [45]. The function of typical CTLs is dependent on Ca^{2+} for inducing microbial agglutination [46]. In the present study, rLvMMR1, as a member of the CTL superfamily, exhibited agglutination activity towards rabbit erythrocytes, two Gram-negative bacteria (*E. coli* and *V. parahaemolyticus*), two Gram-positive bacteria (*B. subtilis* and *S. aureus*), and the fungus (*P. pastoris*) in a Ca^{2+}-dependent manner. Similar functionality has been observed in other MRs. For instance, rPvMR2 was reported to agglutinate *V. parahaemolyticus*, *V. anguillarum*, *V. vulnificus*, and *E. coli* only in the presence of Ca^{2+} [25]. In *P. clarkii*, four recombinant proteins containing CTLDs were all capable of agglutinating bacteria and fungi in presence of Ca^{2+} [24]. Furthermore, the rOnMR protein of *Oreochromis niloticus* could effectively bind and agglutinate *S. agalactiae* and *A. hydrophila* in a Ca^{2+}-dependent manner

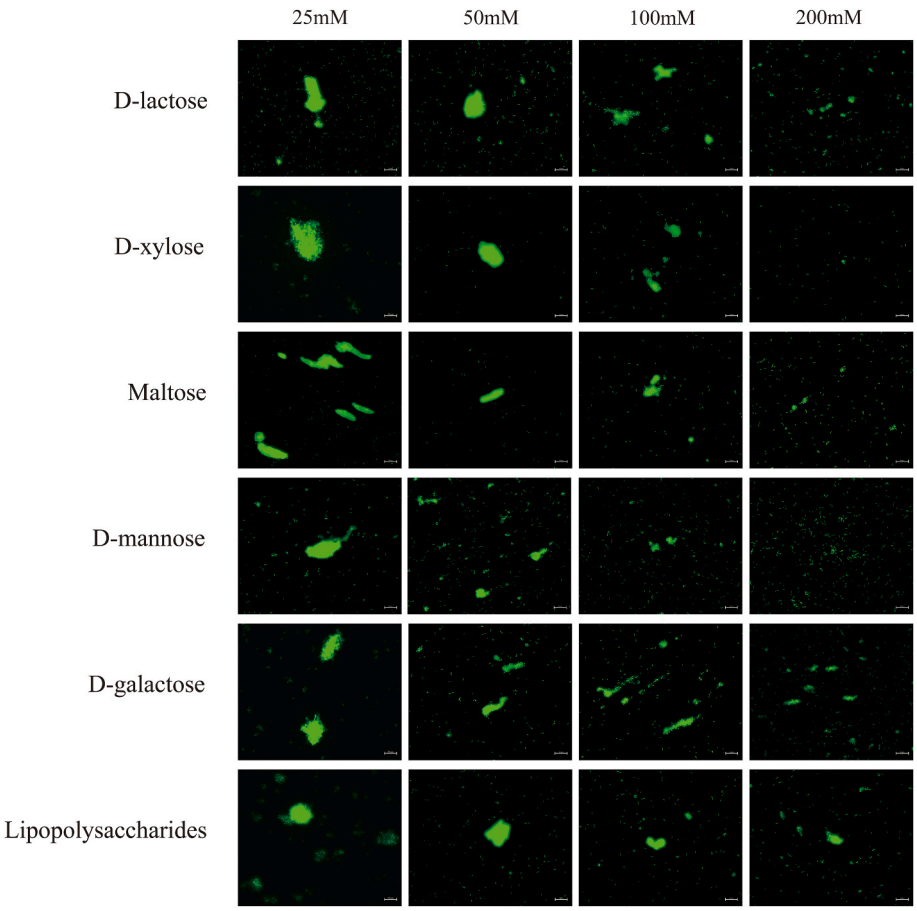


Fig. 7. The carbohydrate binding activity of rLvMMR1. The *Bacillus subtilis* agglutination effects of were inhibited by different concentrations of carbohydrates (25 mM, 50 mM, 100 mM, 200 mM).

[47]. As a member of the CTL superfamily, MR exhibited strong binding affinity for certain components on microbial surfaces via its CTLDs, thereby activating host immunity response [48,49]. In this study, two CTLDs of LvMMR1 contained the carbohydrate-binding signature motifs QPD and EPQ, respectively. Carbohydrate-binding assays demonstrated that D-lactose, D-xylose, maltose, D-galactose, D-mannose, and LPS all bound to rLvMMR1 in a dose-dependent manner, thereby inhibiting its agglutination activity against *B. subtilis*. Similarly, the two CRDs of PvMR2 harbored the characteristic motifs QPT and WHD, and the rPvMR2 protein exhibited the ability to specifically bind polysaccharide molecules [25]. In *F. merguiensis*, FmLC3 contained the characteristic motifs EPN and WYN, and its activity could be inhibited by LPS and mannose [39]. LvMR possessed EPS motif and showed broader carbohydrate recognition specificity, enabling it to bind D-mannose, *N*-acetyl-D-glucosamine and other carbohydrates, such as LPS, D-galactose, D-glucose, *N*-acetylneuraminic acid and *N*-acetyl-β-D-galactosamine [26]. Moreover, even more variant motifs have been identified in CTLs from invertebrates. For example, AiCTL9 from the *Argopecten irradians* contained four CRDs with the motifs YPT, EPD, EPN, and QPN, respectively, and could bind multiple PAMPs, including LPS, PGN, mannan, and glucan [50]. Six CTLs identified in *R. philippinarum* all contained the motifs, EPN and WND [51]. The diversity of carbohydrate-binding motifs in invertebrate CTLs are closely related to their specificity for recognizing carbohydrates on pathogen surfaces, which indicated CTLs including MRs play the important role in innate immunity [18,37].

In conclusion, this study identified a novel MR from *L. vannamei* (LvMMR1), which contained two CTLDs, each harboring the characteristic motif, QPD, LND, EPQ, and FRD, respectively. LvMMR1 was mainly expressed in the hepatopancreas and exhibited a significant immune response to different degrees of EHP infection. The rLvMMR1 performed notable binding and agglutination activity towards rabbit erythrocytes and a broad spectrum of microorganisms. Furthermore, six tested carbohydrates could bind to rLvMMR1 with a dose-dependent manner, simultaneously inhibiting agglutination function of rLvMMR1. Our results demonstrated that LvMMR1, functioning as a CTL, played a crucial role in the immune response of shrimp. These studies suggested the significant role of MR in the immune defense mechanism of shrimp.

Credit author statement

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Data availability

Data will be made available on request.

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